(Please write your Enrollment Number) Enrollment No. 05

End-Term Examination
(CBCS) (SUBJECTIVE TYPE) (Offline)
Course Name: B. Tech CSE/IT/CSEAI/AI&ML, Semester: 3
(November-December, 2023)

Subject Code: BAS 203 | Subject: Numerical Methods

Time: 3 Hours | Maximum Marks: 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV. Perform all the solutions up to the correct four decimal places.

Q1 (2.5*8=20)

(a) Write the number 23.375 in base 2 floating point representation.

(b) Using Newton-Raphson method, locate a real root near 2 of the equation

$$x\log_{10}x - 1.2 = 0.$$

(c) Solve the system of linear equations by Gauss elimination method

$$0.000312x_1 + 0.006032x_2 = 0.003328,$$
$$0.5x_1 + 0.8942x_2 = 0.9471.$$

(d) What is condition number of a matrix? Explain how the condition number is useful in determining whether a matrix is ill-conditioned.

(e) Find the second divided difference of $f(x) = 1/x$, using the points a, b, c.

(f) Using the method of least square fit a straight line to the following data

x	1	2	3	4	5
y	2	4	6	8	10

(g) Given the differential equation

$$\frac{dy}{dx} = x + yx^4$$

with $y(0) = 3$, compute $y(0.1)$ using Picard's method.

(h) Given the initial value problem

$$\frac{dy}{dx} = y - x, y(0) = 2,$$

use RK-method 2th order to obtain $y(0.2)$ when taking $h = 0.1$.

UNIT-I

Q2 Derive secant formula for finding a real root of $f(x) = 0$. Using this formula, determine a root of $x^2e^{-x/2} - 1 = 0$ in the interval $[0, 2]$. The initial values are 1.42 and 1.43. **(5 + 5 = 10)**

Q3 Using General Newton-Raphson method to obtain the double root of the equation $f(x) = x^4 - x^2 - x + 1 = 0$, by taking initial approximation as 0.8. Conclude that the generalized Newton method converges more rapidly than the Newton method. **(8 + 2 = 10)**

UNIT-II

Q4 State the condition of convergence of Successive Over Relaxation (SOR) method. Apply this method, up to four iterations, to solve the system defined by **(2 + 8 = 10)**

	$2x_1 + 17x_2 + 4x_3 = 35,$ $x_1 + 3x_2 + 10x_3 = 24.$ Take $w = 0.5$ and initial approximation vector as $\begin{bmatrix}0\\0\\0\end{bmatrix}$.		
Q5	(a) Define the eigenvalue problem for a square matrix. (b) Using the power method, find the smallest eigen value in magnitude and its corresponding eigenvector of the matrix $A = \begin{bmatrix}2 & -1 & 0\\-1 & 2 & -1\\0 & -1 & 2\end{bmatrix}$, by taking the initial vector as $\begin{bmatrix}1\\1\\1\end{bmatrix}$.	(2 + 8 = 10)	
	UNIT-III		
Q6	Derive Newton's forward difference interpolation formula. Using this formula, find the cubic polynomial which takes the following values $y(1) = 24$, $y(3) = 120$, $y(5) = 336$, and $y(7) = 720$.	(5 + 5 = 10)	
Q7	Evaluate the integral $I = \int_1^2 \frac{2x}{1+x^4} dx$, using (a) Gauss one, two, and three points rules and (b) Simpson's 3/8th rule with $n = 6$.	(5 + 5 = 10)	
	UNIT-IV		
Q8	(a) Using modified Euler's method, obtain solution of the differential equation $\frac{dy}{dx} = x + \sqrt{y}; y(0) = 1$, with step size $h = 0.2$ in the range $x \in [0, 0.4]$. (b) Given the differential equation $\frac{dy}{dx} = x - y^2; y(0) = 1$, use Taylor's series method to determine the value of $y(0.1)$.	(5 + 5 =10)	
Q9	Use RK-method 4th order to find the approximate solution of $y(0.2)$ and $z(0.2)$ as a solution of pair of equations $\frac{dy}{dx} = x + z; \frac{dz}{dx} = y - x$ with the initial conditions $y(0) = 1; z(0) = -1$. Take step size $h = 0.1$.	(10)	

2/2

28th

S-2

350